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Automation in China's Aging Population

Between 1980 and 2015, China's average annual Gross Domestic Product growth exceeded 9% annually (Kuo). This unprecedented growth following the Mao-era enabled China to grow into the second largest global economy, a position it has held since 2010. Much of this growth has been credited to the unique China Model which utilizes China's growing population – increasing by 159.4% between 1950 and 2020 – to drive export-led growth in manufacturing and urban development (Macro Trends). Labor abundance created low wages, and a large population enabled economic growth for decades. However, the one child policy implemented in 1980 alongside rapid urbanization reduced birth rates, creating growing shortages in the manufacturing labor force. As recognized in the Cobb-Douglas Production Function, to maintain constant levels of production with a decreasing labor pool additional capital and/or increases in technology are needed. At the state and policy level, China has recognized the need for advancements in automation to ensure high levels of output. Ultimately, China's aging population will continue accelerating the adoption of automation and robotics in manufacturing as firms substitute capital for shrinking labor supply.

Evidence #1:f

China's labor force historically drove domestic and global growth, being both cheap and incredibly abundant. After a peak of about 1 billion in 2011 (Li et al.), however, the force began to decline, with projections bringing it down as far as around 696 million by 2050. If China's labor force continues on such a sharp decline, labor shortages would worsen, forcing wages up, and ending the cheap labor supply that historically defined China's economy.

The position/seeker ratio, indicating the number of newly added jobs to the market relative to individual job seekers, has been trending upward in China, rising above 1.0 for the first time in 2010 (Li et al). This ratio signals the structural shortage of labor in China.

China rapidly shifted into a low birth rate society, significantly faster than most advanced economies. The U.S. experienced a similar demographic shift over the course of 140 years and the UK in 200. China transitioned over just 30-40 years, however, exhibiting unprecedented speed and magnitude, greatly shifting the demographic make-up of China's population. In 1970, China had a 6 child per woman birth rate. By 2010, this number had declined to 1.4. The most direct cause for this sharp decline was China's "One-Child Policy" enacted in 1979. Limiting every couple to one child, and implementing extreme fines for extra births, this policy was described as "the largest and strictest population control policy in human history" (Banister 1987). However, though clearly a large contributing factor, the "One-Child Policy" was not the only, and not always the main, driver for demographic shift.

Rapid social and economic changes played a critical role in the transition. Even as the government began to relax policies, societal views on birth rates had been shifted, and the norm of "one-child" was instilled upon households. People had been affected by the long-term policy enforcement, as well as other social and economic reforms, and exhibited a shift in societal fertility preferences (Li et al.). Multiple sources note, as well, that across the world, a common trend exists of the decrease in fertility rates alongside the advancement of domestic economies. However it is not the trend that differentiates China's decline, it is, as aforementioned, the speed.

Along with the driving force of the demographic transition, the draining of rural labor pools both shrunk supply and increased cost of labor. The rural to urban influx of labor that once drove economic growth and industrial reform was an abundant source of people, but has since been thoroughly exhausted. By 2007, 98% of rural youth (16-20) were reported to work off the farm (Li et al). This meant that future migrant labor would come at a much higher cost. If recruiting from rural areas, firms and industries would have to employ older workers with established lives and families. Wages would have to increase to incentivize this population and make the move worthwhile.

Evidence #2 Supply Side - Rapid Growth in Robotics:

China's adoption of robots started to support their efforts in becoming a manufacturing powerhouse, competing against the world's most developed economies. Out of developed countries, China has a substantially higher share of employment in manufacturing than all others. From 2001-2012, China did not use much robot labor, and instead relied on their massive labor force. Most of the robots that were utilized came from other countries. In 2012, for example, only 8.7% of the robots sold in China (out of 22,987) were from indigenous firms, indicating China's lack of prowess in designing robots (Freeman et al). In 2013, the Ministry of Industry and Information Technology of People's Republic of China published their first policy promoting the development of robots. In response, local governments began to increase the production, research and development of robots in indigenous firms (Freeman et al). Then, in 2014, China's President Xi Jinping declared a robot revolution in China, saying "Our country will be the biggest market for robots" (Bland). Since then, China has seen a rapid and massive increase in robot development, being the "largest market for robots in the world since 2013" (Freeman et al). As the labor supply shrinks and ages, China needs to replace shrinking labor to retain economic prosperity. We are seeing that robots have been an answer thus far, and for a variety of reasons, we expect this trend to accelerate.

One of the most important reasons why China has the ability to replace labor with robots is because of their educational focus on science and engineering. In order to have enough robots to fill in the gaps left by China's aging population, China needs engineers to design and create robots. It is safe to say China is developing plenty of engineers locally in their education system to fill this need. Data from the U.S. National Science Foundation shows that in every year between 2011 and 2020, China awarded the most science and engineering (S&E) undergraduate degrees of many developed nations. The countries included in this data are: China, Brazil, France, Germany, Mexico, South Korea, the United Kingdom, the United States, India, Japan, and Turkey. Important to this argument, the science and engineering undergraduate degrees earned by Chinese students are heavily concentrated in engineering (the discipline most well suited for robotics), with engineering being 32.8% of the S&E degrees, as opposed to 5.9% in the social and behavioral sciences, and 6.5% in "physical and biological sciences and mathematics and statistics". Data shows a similar trend amongst S&E doctoral degrees. In the same period of 2011-2020, China and the United States awarded the most doctoral degrees in S&E compared to the same countries listed above. China and the US each awarded between 32,000 and 43,000 S&E degrees per year in this period, close to double the amount awarded by all the other countries. Again, China's S&E students focused primarily on engineering, with 36.4% of degrees concentrated in engineering. These trends show China is well positioned to design robots to replace human capital in the labor force and gives China a reason to pursue the robotics opportunity. Further, it would suggest that China is nudging students towards engineering degrees in one of two ways. First, the government and universities could actually be encouraging students to pursue engineering degrees. Second, China could be incentivizing students to choose engineering educational pathways by offering high salaries to workers in engineering related jobs, such as robotics. As Professor Li said in his slides "The Highest Exam," education "channels talented individuals into sectors the government prioritizes." As more qualified engineers enter the workforce, China will be able to accelerate the rate of adoption of automation and robotics in manufacturing and other industries.

Another important supply side factor in China's adoption of robots for manufacturing and labor is investment from the government. Looking just at recent years, there has been massive investment. China is creating a $137 billion fund to support AI and robotics startups (Goh et al) and local governments are making similar investments. Beijing and Guangzhou, for example, launched a $1.4 billion fund to support robotics researchers and designers (Grant-Chapman et al). Given the already substantial investment in robotics from the government, there is reason to believe that China will continue to invest in robotics, making it more likely that China can replace a considerable amount of the labor force with robots. This is because there are already billions of dollars in sunk costs from investments, making it worthwhile to continue investing more in robotics, and making good use of their current investments, rather than wasting money by abandoning their investments and pivoting to a different answer to the shrinking labor supply, such as offshoring labor.

Case Study: Made in China 2025 and Automation Policy

China's governmental investment push is shown very clearly with the 2015 "Made in China 2025" policy implemented. The Chinese government has actively encouraged the adoption of robotics and advanced manufacturing technologies. Launched in 2015 by Prime Minister Li Keqiang, the policy's guiding principles are to replace China's reliance on foreign technology imports with its own innovation. Rather than shifting production abroad in response to rising labor costs, the policy encourages firms to increase productivity through automation within China. It focuses on ten key sectors: new information technology, numerical control tools, aerospace equipment, high-tech ships, railway equipment, energy saving, new materials, medical devices, agricultural machinery, and power equipment (Institute for Security and Development Policy). Most relevant to automation are the information technology and agricultural machinery pushes. This policy served to help China move to become a direct competitor with nations like South Korea, Japan, and Germany. It also addresses China's workforce shrinking capacities, as automation helps maintain industrial output despite declining labor supply due to population aging.

The policy drew inspiration from Germany's Industrie 4.0, which was launched in 2013 and emphasized smart factories, decentralized control, and customized production to enhance competitiveness. They focused on using industry and government collaboration to sustain high-cost manufacturing through automation, while China's policy comes from more of a top-down approach to shift away from low-cost manufacturing.

The Made in China 2025 policy actively pushes automation to make it cheaper than offshoring. The specific goal of "raising domestic content of core components and materials to 40 percent by 2020 and 70 percent by 2025" will particularly help China become more self-sufficient in terms of production capabilities (Institute for Security and Development Policy). The plan focuses on the upgrade of smart manufacturing technology by targeting new generation information technology and high-end computerized machines and robots. Financial support from the government included a semiconductor fund and subsidies from state-owned banks to small and medium sized enterprises. There is also $3 billion available from the Advanced Manufacturing Fund to upgrade technology in key industries.

China's manufacturing sector also benefits from dense industrial clusters and integrated supply chains that make relocation costly. Many Chinese manufacturing hubs, such as those in Guangdong and the Yangtze River Delta, contain networks of specialized suppliers, skilled technicians, logistics infrastructure, and component manufacturers. Moving production to lower-wage countries could disrupt these networks and reduce efficiency. As a result, firms may find it more profitable to invest in automation within China rather than relocate production abroad. In this environment, policies such as Made in China 2025 reinforce incentives to increase productivity through robotics rather than shifting production to cheaper labor markets.

Evidence suggests that automation has accelerated rapidly in China in recent years. According to the International Federation of Robotics, China has become the largest market for industrial robots in the world, accounting for roughly half of global robot installations (IFR International Federation of Robotics). Robot density in Chinese manufacturing has increased dramatically since the mid-2010s, reflecting both rising labor costs and government support for automation. In 2024, domestic market share climbed to 57% last year, up from about 28% over the past decade. These developments underscore the growth of Superstar Firms: firms which have lower labor-to-revenue ratios as they use more capital intensive technologies. Nevertheless, there is still room for continued growth throughout robotics as there is no indication that demand for robots in China will decrease. These trends suggest that policies like Made in China 2025 have contributed to the rapid adoption of robotics across Chinese industry.

China's demographic transition further strengthens these incentives. As the working-age population declines and the dependency ratio rises, firms face increasing pressure to maintain production with fewer workers. Automation provides a way to offset these labor shortages while sustaining manufacturing output. Government policies that encourage robotics adoption therefore interact with demographic trends, reinforcing the shift toward more capital-intensive production.

Why Robots:

As noted above, many of China's core problems in sustaining a strong labor force stems from the One Child Policy alongside other shifting dynamics in the country's population. To understand China's dynamic response to the shrinking population through its investment in robots, many clear indicators reveal why this transition is necessary.

First, many reports reveal offshoring is not a realistic alternative for China's unique manufacturing industry. The Rhodium Group reports, "Today, no country can replicate China's highly optimized production ecosystem at scale, so firms remain slow to relocate to alternative production hubs, even in lower-tech, lower-value-add sectors such as apparel" (Kratz et al). The unique structures of China's key businesses reinforce how China became a global leader in exports over the past 50 years: they integrate highly efficient processes which are hard to replicate. In part, this growth stemmed from investment to offshore work *to* China because of the abundance of cheap labor; reassigning offshore capabilities to other countries proves difficult given the complexity of China's manufacturing economy. Additionally, given the rapid increase of manufacturing wages domestically, the financial incentive to offshore production is shrinking given the high relocation costs. Nevertheless, while some offshoring has occurred to other

Southeast Asian countries, those economies remain too small, infrastructure limited, lack sufficient port infrastructure, and face increasingly expensive production.

Additionally, given China's complex geopolitical relationships, President Xi introduced China's Dual Circulation Strategy (DCS) in the early 2020s. DCS arose following pandemic-related social and economic disruptions, underscoring the importance of self-sufficiency and national security. DCS aims to prioritize internal circulation – domestic consumption, services, and innovation – while relying on external circulation such as trade and investment to support growth. In implementing the DCS, President Xi affirmed China's commitment to self-reliance, creating a resilient and unique development model.

To achieve continued economic growth and self-reliance, the State has introduced several policies to pursue advancement as noted above. Many policies advocate utilizing capital and technology (robots and automation) to serve as a direct substitute for labor. Recognizing the natural transition which can occur from labor to robots, the Chinese government allocated over $20b in subsidies to the robotics industry in the form of grants, loans, tax, credits, and state-backed venture capital funding to help throughout 2024 (Grant-Chapman). The success of similar programs is highlighted in recognizing that 60% of China's robot installments are produced domestically, driving costs farther down than anywhere else and further making automation cheaper relative to offshoring (Szyzdek). The additional subsidies supplement the Made in China 2025 policy, recognizing a government push to integrate robotics into the operations of the modern economy. These subsidies have been implemented alongside the explosion of rigorous STEM education throughout China, providing the necessary human capital to build and deploy robots throughout the manufacturing economy.

Limitations:

Although automation is increasing rapidly in China, similar trends are occurring across the global economy and China's desire to keep up also contributes to their push for automation. Additionally, in other countries, an aging population cannot be the reason for their automation. Across the world, a major reason for the drive for automation is credited to rising labor costs and not demographic change. China's manufacturing wages have increased dramatically over the last two decades, reducing the country's historical labor cost advantage. When wages increase, firms may substitute capital for labor regardless of demographic trends.

The paper "*Robots and Jobs: Evidence from US Labor Markets"* by Daron Acemoglu and Pascual Restrepo, featured in the University of Chicago's Journal of Political Economy, Volume 128 and published in June 2020, provides strong evidence that automation responds to economic incentives rather than demographic factors alone. Using U.S. labor market data, the authors find that the introduction of one additional industrial robot per 1,000 workers reduces wages by approximately 0.42 percent and displaces workers in affected industries (Acemoglu and Restrepo). Their findings demonstrate that firms adopt robotics when doing so lowers production costs relative to labor. While the study focuses on the United States, the economic logic applies more broadly: when labor becomes more expensive, firms substitute capital for labor. In China, manufacturing wages have more than tripled between 2005 and 2020 as the country has developed economically (Glennen). Rising wages therefore reduce China's historical advantage of cheap labor and increase incentives for automation regardless of demographic trends.

Global manufacturing competition also drives automation growth in China. Richard Baldwin's book *The Great Convergence* highlights the role of global competition in driving technological adoption. Baldwin argues that globalization and advances in communication technology have allowed manufacturing knowledge and production techniques to spread rapidly across countries. As a result, firms increasingly compete on productivity, technological capability, and efficiency rather than solely on labor costs. In this environment, countries that fail to adopt advanced manufacturing technologies risk losing competitiveness in global supply chains. As China continues to advance its manufacturing capabilities, firms must compete with technologically advanced economies such as Germany, Japan, and South Korea that have already adopted high levels of automation in manufacturing. Specifically, as of 2017, Germany averages more than 300 industrial robots per 10,000 industry employees, while China only averages 19 (Baldwin). Closing this gap has become an important objective for Chinese manufacturers seeking to remain competitive in international markets. Since 2017, China has closed this gap to now have 470 robots per 10,000 workers, showing the recent strength of the push of their policy and push for competitiveness in global markets (International Federation of Robotics).

Another motivator for technological progress in China is technological progress itself. Advances in robotics and artificial intelligence have lowered the cost of automation while improving the capabilities of robotic systems. As these technologies become more affordable and efficient, firms may adopt them regardless of demographic trends. Graetz and Michaels analyzed the impact of industrial robots across multiple advanced economies in *"Robots at Work*," featured in the 2018 Review of Economics and Statistics, and found that robot adoption raised total factor productivity and lowered output prices. Their research estimates that robotics contributed approximately 0.36 percentage points to annual GDP growth across the 17 countries studied. As robotics technology becomes more efficient and affordable, firms may adopt automation even without strong demographic pressures. Improvements in artificial intelligence, machine vision, and robotics engineering have reduced the cost of automation while expanding the range of tasks robots can perform. As a result, technological progress may independently drive automation adoption in China regardless of demographic trends.

Another limitation to the demographic explanation for automation is the significant role of government industrial policy. China has pursued an active industrial strategy aimed at promoting technological upgrading in manufacturing sectors. Programs such as Made in China 2025 provide subsidies, tax incentives, and research funding to encourage firms to adopt automation technologies. These policies may stimulate robot adoption even in the absence of demographic pressures. In this sense, the rise of automation in China may reflect deliberate government intervention designed to increase technological capabilities and global competitiveness in the wake of population aging and productivity losses rather than solely a response to population aging.

Another limitation to the argument is that automation does not always fully replace human labor. In many industries, robots and automation technologies complement workers rather than substitute for them entirely. Graetz and Michaels (2018) find that robot adoption increases productivity and economic output across advanced economies, but does not necessarily eliminate large numbers of jobs overall. Instead, automation can change the types of tasks workers perform, shifting labor toward roles that require supervision, maintenance, and technical skills. In the Chinese context, this suggests that robots may not completely offset the effects of a shrinking labor force. Even with increased automation, firms may still require significant human labor, particularly in complex or specialized production processes.

However, China's manufacturing ecosystem differs from that of many Western economies. China possesses dense industrial clusters, extensive supplier networks, and highly developed infrastructure that support large-scale manufacturing stemming from other countries offshoring to China as noted above. Relocating production abroad could disrupt these networks and reduce efficiency. As a result, firms may find it more advantageous to automate production within China rather than shift manufacturing overseas.

While rising wages, global competition, technological progress, and industrial policy all contribute to the adoption of automation, demographic aging amplifies these forces by increasing the urgency of maintaining production with a shrinking labor force. In China's case, population aging interacts with these broader economic dynamics to accelerate the shift toward more capital-intensive manufacturing. Rather than acting as the sole cause of automation, demographic change strengthens existing incentives for firms to invest in robotics and advanced manufacturing technologies. As labor becomes scarcer and more expensive, automation becomes an increasingly attractive strategy for maintaining productivity and economic growth.

Conclusion & Predictions:

China's economic growth has historically relied on an abundant and low-cost labor force. However, rapid demographic change driven by declining fertility and population aging is fundamentally altering this model. As the working-age population shrinks, labor shortages and rising wages are reducing the viability of labor-intensive production. In response, firms are increasingly investing in automation and robotics as a substitute for human labor. Evidence from rising robot adoption, government investment, and China's growing base of engineering talent all suggest that this transition is already underway.

While automation in China is influenced by multiple factors, including rising wages, global competition, technological progress, and industrial policy, demographic aging plays a critical role in accelerating these trends. Rather than acting as the sole driver, population aging amplifies existing economic incentives by increasing the urgency of maintaining output with a smaller labor force. In this sense, China's demographic transition reinforces the shift toward more capital-intensive production.

Looking forward, several key trends are likely to shape the future of China's economy. First, the adoption of robotics and automation in manufacturing is expected to continue accelerating as labor becomes increasingly scarce and costly. Second, China's production model will become more capital-intensive, with firms relying more heavily on technology and less on manual labor. Finally, automation will play a central role in sustaining productivity and supporting long-term economic growth despite demographic headwinds. As China navigates this transition, its ability to successfully integrate automation into its manufacturing sector will be critical in determining whether it can maintain its position as a global economic leader.

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